A Major Project Report

on

AI BASED CAREER AND SKILL RECOMMENDATION SYSTEM

submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE & ENGINEERING

By

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CERTIFICATE

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DECLARATION BY THE CANDIDATE

We, hereby declare that the project report entitled "**AI BASED CAREER AND SKILL RECOMMENDATION SYSTEM**", under the guidance of **Dr. Raziuddin, Head of the Department,** Department of Computer Science and Engineering, **Lords Institute of Engineering & Technology,** affiliated to **Osmania University, Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering.**

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree.

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Lords Institute of Engineering and Technology strives for excellence in professional education through quality, innovation and teamwork and aims to emerge as a premier institute in the state and across the nation.

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- To provide a scholarly and vibrant learning environment that enables faculty, staff and students to achieve personal and professional growth.
- To contribute to advancement of knowledge, in both fundamental and applied areas of engineering and technology.
- To forge mutually beneficial relationships with government organizations, industries, society and the alumni.

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DM2: Develop design, research, and entrepreneurial skills for successful career

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B.E. Computer Science and Engineering Program Educational Objectives (PEOs):

PEO1	Exhibit strong foundations in Basic Sciences, Computer Science and allied engineering
PEO2	Identify, formulate, analyze and create professional solutions, novel products using appropriate tools and techniques, design skills.
PEO3	Pursue successful career with continuous learning, with emphasis on competency oriented towards industry
PEO4	Practice ethics, managerial and leadership skills to work cohesively within a group

Head of the Department
Computer Science and Engineering
Lords Institute of Engg & Tech
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B.E. Computer Science and Engineering Program Outcomes (POs):

Engineering Graduates will be able to:

S. No.	Program Outcomes (POs)
1.	**PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2.	**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	**PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	**PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	**PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7.	**PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	**PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9.	**PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	**PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11.	**PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

B.E. Computer Science and Engineering Program Specific Outcomes (PSO's):

PSO1	**Professional Skills:** Implement computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics and networking for efficient analysis and design of computer-based systems of varying complexity
PSO2	**Problem-Solving Skills:** Apply standard practices and strategies in software service management using open-ended programming environment with agility to deliver a quality service for business success

Head of the Department
Computer Science and Engineering
Lords Institute of Engg & Tech
Hyderabad - 500091. T.S.

Course Outcomes: C424 - Major Project

Student will be able to

CO. No	Description	Blooms Taxonomy Level
C424.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to real-world problems.	
C424.2	Evaluate different solutions based on economic and technical feasibility.	
C424.3	Effectively plan a a project and confidently perform all aspects of project management	
C424.4	Demonstrate effective oral communication skills	
C424.5	Demonstrate effective team work and written skills	

Course Articulation Matrix:

Mapping of Course Outcomes (CO) with Program Outcomes (PO) and Program Specific Outcomes (PSO's):

Course Outcome s (CO)	Program Outcomes (PO)											Program Specific Outcomes (PSO's)	
	PO 1	PO 2	PO3	PO4	PO5	PO6	PO 7	PO 8	PO9	PO1 0	PO1 1	PSO 1	PSO 2
C424.1.													
C424.2.													
C424.3.													
C424.4.													
C424.5.													
Average													

Level:

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

<u>SDG Mapping:</u>

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE		15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

ABSTRACT

In the era of digital recruitment and rapidly evolving skill demands, traditional career guidance systems fail to provide personalized, data-driven support for job seekers. This paper proposes **AI based Career and Skill Recommendation System**, an intelligent and integrated career development platform that leverages advanced Artificial Intelligence and Natural Language Processing techniques to assist users across the entire career lifecycle. The system combines multiple functionalities, including dynamic career assessment, AI-based resume and cover letter generation, adaptive mock interview simulation, skill gap analysis, and location-aware job search, within a single unified framework. Drawing insights from existing research on AI resume screening, interview systems, and job matching algorithms, the proposed solution enhances accuracy through semantic understanding and contextual analysis rather than simple keyword matching. The system also employs efficient caching mechanisms and scalable architecture to ensure high performance and responsiveness. Experimental observations indicate significant improvements in user engagement, career alignment, and preparation efficiency compared to traditional methods. The proposed approach not only reduces the complexity of career planning but also provides a scalable and intelligent solution to bridge the gap between academic learning and industry expectations.

TABLE OF CONTENTS

CONTENTS	PAGE NUMBERS
Title	i
Certificate	ii
Declaration	iii
Acknowledgement	iv
Vision & Mission of the Institute	v
Vision & Vision of the Department	vi
Program Educational Objectives (PEOs)	vii
Program Outcomes (POs)	viii-ix
Program Specific Outcomes (PSOs)	x
Course Outcomes	xi
Course Articulation Matrix	xii
SDG Mapping	xii
Abstract	xiii
Table Of Content	xiv-xvi
List Of Figures	xvii
List Of Tables	xviii
CHAPTER 1	
INTRODUCTION	1-5
1.1 Introduction	1
1.2 Objectives	1-2
1.3 Problem Formulation	2-3
1.4 Existing System	3-4
1.5 Proposed System	4-5
1.6 Unique Features of the System	5
CHAPTER 2	
LITERATURE SURVEY	6-7
CHAPTER 3	
REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION	8-12
3.1 Requirement Analysis	8-10
3.1.1 User Requirements	8

3.1.2 System Requirements 9
3.1.3 Functional Requirements 9-10
3.1.4 Non-Functional Requirements 10
3.2 System Specification 10-12
3.2.1 Hardware Specification 10
3.2.2 Software Specification 10-11
3.2.3 System Architecture Specification 11
3.2.4 Data Flow Specification 12
3.2.5 Constraints 12
CHAPTER 4
SYSTEM DESIGN 13-26
4.1 Overview of System Design 13
4.2 Architectural Design 13-16
4.3 Module Design 16-20
4.4 Data Flow Design 20
4.5 Database Design 20-21
4.6 Interface Design 21-22
4.7 Security Design 22
4.8 Performance Design 22-23
4.9 Algorithmic Design 23
4.10 File Structure Design 23
4.11 UML Diagrams 24-26
4.11.1 Use Case Diagram Theory 24
4.11.2 Activity Diagram 24-25
4.11.3 Sequence Diagram 25
4.11.4 Class Diagram Theory 26
CHAPTER 5
IMPLEMENTATION 27-35
5.1 Methodolgies 27-28
5.2 Algorithms 28-29
5.3 System Implementation Details 29-30
5.4 Sample Code 31-35
CHAPTER 6

TESTING 36-41
6.1 Types of Testing 36-38
6.2 Test Cases 38-41
CHAPTER 7
RESULTS 42-46
7.1 Overview of Results 42
7.2 Functional Results 42-43
7.3 User Experience Results 43
7.4 Accuracy And Efficiency 44
7.5 Comparative Results 44-46
CHAPTER 8
CONCLUSION AND FUTURE SCOPE 47-48
8.1 Conclusion 47
8.2 Future Scope 47-48
REFERENCES 49

LIST OF FIGURES

Figure Number	Name of the Figure	Page Number
4.1	Use Case UML Diagram	24
4.2	Sequence UML Diagram	25
4.3	Class UML Diagram	26
7.1	Landing Page	45
7.2	Career Result Page 1	45
7.3	Career Result Page 3	46
7.4	Internship Recommendation	46

LIST OF TABLES

Table Number	Table Title	Page Number
1.1	Objectives of the System	2
1.3	Features of Proposed System	4-5
5.1	Algorithms used in the System	29
6.1	Test Case 1-User Registration	38
6.2	Test Case 2-User Login	39
6.3	Test Case 3-Career Assessment	39
6.4	Test Case 4-Resume Generation	40
6,5	Test Case 5-Mock Interview	40
6.6	Test Case 6-Job Research	41
6.7	Test Case 7-Performance Check	41
7.1	Comparative Analysis of Existing System and Proposed System	44

CHAPTER 1
INTRODUCTION

1.1 Introduction

In today's rapidly evolving global job market, individuals face significant challenges in identifying suitable career paths, aligning their skills with industry demands, and effectively presenting themselves to potential employers. The transition from academic education to professional life, as well as career shifts across industries, often leads to confusion, stress, and poor decision-making due to a lack of proper guidance and real-time insights.

Traditional career guidance systems are limited by static information, lack of personalization, and minimal integration with modern recruitment processes such as Applicant Tracking Systems (ATS). Additionally, job seekers frequently struggle with resume optimization, interview preparation, and identifying relevant opportunities, resulting in inefficient career progression.

To address these challenges, the **AI Based Career and Skill Recommendation System** is introduced as an advanced, AI-driven career management and optimization platform. This system leverages modern technologies such as Large Language Models (LLMs), specifically advanced AI architectures, to provide intelligent, personalized, and data-driven career guidance.

AI Based Career and Skill Recommendation System acts as a virtual career counselor that assists users in career discovery, skill gap analysis, roadmap generation, resume building, interview preparation, and job searching. By integrating multiple functionalities into a single platform, the system provides a seamless and efficient solution for career development.

1.2 Objectives

The primary objectives of the AI Based Career and Skill Recommendation System are as follows:

- To develop an intelligent system that analyzes user skills, interests, and experience to recommend suitable career paths.

- To provide personalized, data-driven career guidance using advanced AI technologies.
- To automate the creation of ATS-friendly resumes and customized cover letters.
- To generate structured learning roadmaps (3, 6, and 12 months) based on skill gap analysis.
- To simulate real-time mock interviews with dynamic question generation and feedback.
- To provide a fast and efficient job and internship search system based on user location and preferences.
- To create a unified platform that integrates career discovery, preparation, and application processes.
- To continuously adapt recommendations based on user progress and market trends.

S.No	Objective	Description
1	Career Recommendation	Suggest suitable career paths using AI
2	Resume Generation	Create ATS-friendly resumes
3	Interview Preparation	Provide mock interview system
4	Skill Analysis	Identify skill gaps
5	Job Search	Provide location-based job results

Table 1.1: Objectives of the System

1.3 Problem Formulation

The modern job market suffers from a significant gap between candidate capabilities and industry requirements. Many individuals possess the necessary knowledge and skills but fail to secure opportunities due to poor career direction, ineffective presentation, and lack of preparation.

Key problems identified include:

- Lack of personalized career guidance systems for both freshers and experienced professionals.

- Difficulty in identifying suitable career paths due to information overload and market complexity.
- Ineffective resumes that fail to pass ATS filters.
- Absence of structured learning plans tailored to individual skill gaps.
- Limited access to realistic and affordable interview preparation tools.
- Fragmentation of tools required for career development (resume builders, job portals, learning platforms, etc.).
- Lack of real-time feedback and adaptive career planning mechanisms.

These challenges result in unemployment, underemployment, and increased stress among job seekers, highlighting the need for an integrated and intelligent solution.

1.4 Existing System

The existing career development ecosystem consists of multiple independent platforms such as job portals, resume builders, and learning platforms.

- **Job Portals (e.g., LinkedIn, Indeed):** Provide job listings but lack personalized career guidance and skill analysis.
- **Resume Builders:** Focus mainly on design templates without deep understanding of job requirements or ATS optimization.
- **Learning Platforms (e.g., Coursera, Udemy):** Offer courses but do not tailor learning paths based on individual needs.
- **Mock Interview Platforms:** Often expensive and limited to specific domains.

Disadvantages of Existing System:

- Lack of integration between different career development stages.
- Minimal personalization and adaptive intelligence.
- No proper career discovery mechanism.

- Inefficient job matching and application process.
- High dependency on user knowledge and manual effort.

1.5 Proposed System

The proposed system, **AI Based Career and Skill Recommendation System**, is a unified, AI-driven platform designed to overcome the limitations of existing systems.

It integrates multiple functionalities into a single ecosystem, including:

- AI-based career assessment and recommendation
- Skill gap analysis and roadmap generation
- ATS-optimized resume and cover letter creation
- AI-powered mock interview system
- Location-based job and internship search

The system utilizes advanced AI models to analyze user inputs and generate highly personalized outputs. It also incorporates modern web technologies and serverless architecture to ensure high performance, scalability, and reliability.

Advantages of Proposed System:

- Fully personalized career guidance
- Integration of all career development stages in one platform
- Real-time feedback and adaptive recommendations

Feature	**AI Based Career and Skill Recommendation System**
Career Guidance	AI-based personalized recommendations
Resume Builder	ATS-optimized dynamic resumes

Feature	AI Based Career and Skill Recommendation System
Interview System	AI-based adaptive interviews
Job Search	Fast, cached, location-based
Integration	All features in one platform

Table 1.3: Features of Proposed System

1.6 Unique Features of the System

The AI Based Career and Skill Recommendation System offers several unique features that distinguish it from traditional systems:

- **AI-Powered Career Assessment:** Dynamically analyzes user skills, interests, and experience to suggest optimal career paths.
- **Adaptive Roadmap Generation:** Provides personalized learning plans based on individual skill gaps and timelines.
- **ATS-Optimized Resume Builder:** Automatically generates resumes that align with modern recruitment systems.
- **Dynamic Mock Interview System:** Generates role-specific questions and provides real-time feedback and scoring.
- **Integrated Career Ecosystem:** Combines discovery, preparation, and job search into a single platform.
- **Context-Aware Intelligence:** Uses AI to understand user background and provide highly relevant recommendations.
- **Efficient Job Search with Caching:** Reduces latency and improves performance through intelligent caching mechanisms.

CHAPTER 2
LITERATURE SURVEY

A thorough and critical review of the existing landscape for career development tools, educational platforms, and job portals reveals several primary categories of systems. Each of these categories exhibits distinct strengths but also significant limitations that AI Based Career and Skill Recommendation System specifically aims to resolve through its integrated architecture.

General Career and Networking Platforms like LinkedIn, Indeed, and Glassdoor are undeniably the industry standard for job aggregation, employer branding, and professional networking. However, they operate fundamentally on a pull-based, search-oriented model. The user must already possess a clear understanding of what they are looking for (e.g., job titles, specific companies) to extract value. They offer virtually no algorithmic guidance for individuals in the 'discovery phase' who are unsure of their optimal career path. Furthermore, they do not provide the granular, personalized, AI-driven feedback necessary to refine a candidate's technical profile, improve their resume format for ATS algorithms, or prepare them for the behavioral and technical rigor of specific job interviews.

Automated Resume and CV Builders like Canva, Zety, and Novoresume excel at their primary function: providing aesthetically pleasing, graphically rich templates for resumes. However, from a technical standpoint, they are often "AI-lite." They may provide basic spell-checking or generic, pre-written bullet points for common roles, but they fail to understand the deep semantic context of a user's unique experience relative to a specific, nuanced job description. They prioritize visual design over ATS compatibility (many graphical templates are unreadable to ATS parsers) and do not bridge the gap between building a document and preparing for the actual interview process that the document is meant to secure.

Learning Management Systems and Static Roadmaps like roadmap.sh, Coursera, and Udemy provide incredible community-driven, highly detailed visual paths for learning various technological roles (e.g., Full-Stack Developer, DevOps Engineer, Data Scientist). However, these roadmaps are static, linear, and universal. They assume every

learner is starting from scratch. They do not account for a user's existing knowledge base, their prior transferrable skills, their specific required timeline (e.g., needing to find employment in 3 months versus having 12 months to study), or their geographic location.

AI Mock Interview Platforms like Interviewing.io and Pramp are highly effective for their specific niche, but they are often prohibitively expensive, intimidating for beginners or freshers, and overly focused on very high-level software engineering roles involving complex data structures and algorithms. They leave individuals in other fields, or those seeking entry-level positions, without adequate, accessible, and less intimidating support mechanisms for interview practice.

AI Based Career and Skill Recommendation System fundamentally differentiates itself by synthesizing the strengths of these disparate categories while systematically eliminating their weaknesses through the application of advanced AI. It uses Google Gemini 2.0 Flash to perform Deep Semantic Analysis of resumes to ensure ATS compliance, Sentiment and Logical Analysis of interview answers to provide actionable feedback, and Predictive Modeling based on complex multidimensional assessments to recommend optimal career paths. It is the first platform in its class to offer a truly circular, interconnected feedback loop: Assessment -> Personalized Learning Roadmap -> AI-Optimized Documentation -> Realistic Practice -> Job Discovery.

CHAPTER 3

REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

3.1. Requirement Analysis

Requirement Analysis is a critical phase in the development of the **AI Based Career and Skill Recommendation System**, where user needs, system functionalities, and operational constraints are identified and clearly defined. This phase ensures that the system is designed to effectively address real-world problems related to career guidance, job preparation, and skill development.

3.1.1 User Requirements

The primary users of the system are:

- Students (Freshers)
- Job seekers
- Experienced professionals

The system must satisfy the following user needs:

- Provide **personalized career recommendations** based on skills, interests, and experience.
- Generate **ATS-optimized resumes and cover letters**.
- Offer **career roadmaps** with clear timelines (3/6/12 months).
- Enable **mock interview practice** with AI feedback.
- Provide **real-time job and internship search** based on location.
- Maintain a **centralized dashboard** for tracking progress and updates.

3.1.2 System Requirements

The system must perform the following core operations:

- Collect and process user data through dynamic assessments.
- Analyze user inputs using **AI (Google Gemini 2.0 Flash)**.
- Generate personalized outputs such as:
  - Career recommendations
  - Skill gap analysis
  - Learning roadmap
  - Resume and cover letter
- Store and manage user data securely using a database.
- Integrate external APIs for job search functionality.
- Ensure smooth interaction between frontend and backend components.

3.1.3 Functional Requirements

The system shall:

- Authenticate users securely using Clerk.
- Categorize users as **Fresher or Experienced**.
- Conduct adaptive questionnaires.
- Generate **top 3 career recommendations** with reasoning.
- Create **AI-generated resumes and cover letters**.
- Provide **mock interview simulations with scoring**.
- Generate **time-bound learning roadmaps**.

- Fetch **location-based job and internship listings**.
- Allow users to update profiles and track progress.

3.1.4 Non-Functional Requirements

The system must ensure:

- **Performance:** Fast loading (TTI < 2.5 seconds).
- **Scalability:** Serverless architecture (Vercel + Neon DB).
- **Security:** Encrypted communication (SSL/TLS).
- **Reliability:** Background task handling via Inngest.
- **Usability:** Mobile-first responsive design.
- **Maintainability:** Modular and reusable code structure.

3.2. System Specification

System Specification defines the technical environment and constraints under which the system operates.

3.2.1 Hardware Specification

The minimum hardware requirements are:

- Processor: Dual-Core CPU (Intel i5 / Apple M1 or equivalent)
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 10 GB free space (SSD preferred)

3.2.2 Software Specification

The system requires the following software environment:

- Operating System: Windows / macOS / Linux
- Programming Languages: JavaScript / TypeScript

- Framework: Next.js 15 (React 19)
- Database: PostgreSQL (Neon DB)
- ORM: Prisma
- AI Engine: Google Gemini 2.0 Flash
- Authentication: Clerk
- Styling: Tailwind CSS + Shadcn UI

3.2.3 System Architecture Specification

The system follows a **Serverless Client-Server Architecture**, consisting of:

1. **Frontend Layer**
   - Built using React (Next.js)
   - Handles user interaction and UI rendering
2. **Backend Layer**
   - Next.js Server Actions
   - Handles business logic and API communication
3. **Database Layer**
   - PostgreSQL via Prisma ORM
   - Stores user profiles, assessments, and outputs
4. **AI Integration Layer**
   - Google Gemini API for intelligent processing
5. **External API Layer**
   - JSearch API for job and internship data

3.2.4 Data Flow Specification

The system processes data in the following sequence:

1. User inputs data through UI.
2. Data is validated and sent to server actions.
3. Server communicates with:
   - Database (Prisma + Neon)
   - AI engine (Gemini API)
4. AI processes and generates output.
5. Results are stored and displayed to the user.

3.2.5 Constraints

The system operates under the following constraints:

- Dependency on external APIs (Gemini, JSearch)
- Internet connectivity required
- API rate limits and cost considerations
- Serverless execution limits (Vercel)

CHAPTER 4

SYSTEM DESIGN

The **System Design** of **AI Based Career and Skill Recommendation System** is centered on creating an intelligent, scalable, secure, and user-friendly career management ecosystem. The platform is designed as a **full-stack AI-driven web application** that integrates user assessment, AI-based recommendation, professional document generation, interview preparation, and job discovery into a single unified system. The design ensures that all modules work together smoothly while maintaining high performance, modularity, and security.

4.1. Overview of System Design

AI Based Career and Skill Recommendation System follows a **serverless-oriented modular design**. Instead of building separate disconnected systems for career guidance, resume building, interview preparation, and job search, the platform unifies all these functions inside one centralized architecture. The design focuses on the complete user journey, beginning from self-assessment and ending with employment readiness.

The system is designed around the following major principles:

- **User-centric personalization**
- **AI-driven decision making**
- **Modular full-stack architecture**
- **Secure data handling**
- **Scalable cloud-native deployment**
- **Fast response through caching and optimized API design**

4.2. Architectural Design

The overall architecture of AI Based Career and Skill Recommendation System can be divided into five major layers:

a) Presentation Layer

This is the front-end layer through which users interact with the platform. It is built using **Next.js 15 with React 19**, supported by **Tailwind CSS** and **Shadcn UI** for responsive and modern user interface design.

This layer is responsible for:

- Rendering onboarding pages
- Displaying assessment forms
- Showing recommended career paths
- Providing resume and cover letter editors
- Conducting mock interviews
- Displaying roadmaps and job listings
- Maintaining dashboard interaction

The front-end is designed to be **mobile-first and responsive**, allowing access from smartphones, tablets, and desktops.

b) Application Logic Layer

The application logic is handled by **Next.js Server Actions**. This layer acts as the secure bridge between the client-side interface and the backend services. All sensitive logic such as AI prompting, database access, and API communication is executed here.

This layer is responsible for:

- Processing user assessment data
- Calling Gemini AI for analysis
- Generating resumes, cover letters, and roadmaps
- Managing job search requests

- Handling mock interview evaluation
- Enforcing business rules and workflow logic

c) Intelligence Layer

The intelligence layer is the core innovation of the system. It uses **Google Gemini 2.0 Flash** to perform context-aware analysis and generation tasks.

It is responsible for:

- Career recommendation generation
- Skill gap analysis
- Resume optimization
- Cover letter synthesis
- Mock interview question generation
- AI-based answer feedback
- Learning roadmap generation

This layer gives the system its adaptive and personalized nature.

d) Data Persistence Layer

The persistent storage layer uses **PostgreSQL through Neon DB** with **Prisma ORM**. This layer stores all user-related and system-related information in a structured relational model.

It is responsible for storing:

- User profile data
- Assessment responses
- Recommended careers
- Resume content

- Cover letters
- Roadmaps
- Interview history
- Industry insights
- Feedback records

Prisma ensures type-safe database access and helps maintain relational consistency.

e) External Integration Layer

This layer handles communication with third-party services such as:

- **Clerk** for authentication and authorization
- **Google Gemini API** for AI intelligence
- **JSearch API / RapidAPI** for internships and jobs
- **Inngest** for background task execution

This external layer expands the capabilities of the platform without overloading the internal architecture.

4.3. Module Design

The system is divided into several major modules, each designed to perform a specific function.

a) Authentication Module

This module manages user registration, login, and secure session handling using **Clerk**.

Functions:

- User sign-up and sign-in
- Session persistence

- Protected route handling
- Identity management

b) User Assessment Module

This module gathers data about the user's skills, experience, interests, and goals. The design is dynamic and adaptive.

For **Freshers**, the module focuses on:

- Interests
- Academic background
- Psychological inclinations
- Latent skills

For **Experienced users**, it focuses on:

- Past work experience
- Existing technical stack
- Transition goals
- Career growth direction

This module passes structured input to the AI engine for career analysis.

c) Career Recommendation Module

This module processes assessment inputs through Gemini and generates the **top three recommended career paths**. Each recommendation includes:

- Role title
- Match reasoning
- Industry fit

- Skill relevance
- Possible location relevance

This is one of the central decision-support modules in the system.

d) Resume Builder Module

This module converts raw user information into a professional, **ATS-friendly resume** in clean Markdown format.

Functions:

- Resume content generation
- Bullet point refinement
- Achievement-oriented writing
- ATS optimization

e) Cover Letter Generator Module

This module merges user profile data with a pasted job description and creates a tailored cover letter.

Functions:

- Contextual JD analysis
- Alignment with user background
- Professional formatting
- Personalized content creation

f) Skill Gap and Roadmap Module

This module compares current user skills against industry requirements and generates a roadmap for improvement.

Functions:

- Skill gap identification
- 3-month roadmap generation
- 6-month roadmap generation
- 12-month roadmap generation
- Milestone creation
- Progress tracking suggestions

g) Mock Interview Module

This module simulates interviews and provides intelligent feedback.

Functions:

- Dynamic question generation
- Technical and behavioral question handling
- Answer evaluation
- Scoring and feedback
- Progressive increase in difficulty

h) Job and Internship Search Module

This module fetches relevant jobs and internships based on user location and target career path.

Functions:

- Local and remote listing retrieval
- External API calls
- Query filtering by city/country
- Cached result display

i) Feedback Module

This module collects feedback from users about recommendation quality and system usefulness. It helps future evaluation and model improvement.

4.4. Data Flow Design

The system follows a structured data flow:

1. The user enters the platform and logs in.
2. The user completes onboarding and assessment.
3. The client sends the data to the server action.
4. The server validates the request and fetches any required user context from the database.
5. A carefully structured prompt is generated and sent to Gemini AI.
6. Gemini returns the analyzed output such as recommended roles, roadmap, or interview questions.
7. The server stores important results in Neon DB via Prisma.
8. The frontend retrieves and displays the result in an interactive format.
9. For job search, the server first checks the cache. If a valid cached response exists, it is returned immediately. Otherwise, the external API is queried and the result is cached.

This flow ensures both **accuracy and efficiency**.

4.5. Database Design

The database design follows a relational structure with the **User** entity as the central table. Other tables are linked to the user in one-to-many or related relationships.

Typical entities include:

- User

- Resume
- Assessment
- CareerRecommendation
- Roadmap
- InterviewQuiz
- Feedback
- IndustryInsight

This design ensures:

- structured storage
- reduced data duplication
- easy querying
- data integrity through foreign keys

4.6. Interface Design

The interface design emphasizes clarity and ease of use. The dashboard acts as the central point of access to all major features.

Key interface elements include:

- onboarding screens
- assessment forms
- recommendation cards
- markdown editor
- roadmap panels
- interview answer input areas

- internship and job listing sections

The design uses reusable UI components so that visual consistency is maintained throughout the platform.

4.7. Security Design

Security is a major part of the system design. The system is designed so that sensitive operations are never directly exposed to the client side.

Security measures include:

- authentication through Clerk
- protected server-side actions
- encrypted communication using SSL/TLS
- scoped database access using Prisma
- secure handling of API keys
- user-specific data isolation

This ensures that users can only access and modify their own records.

4.8. Performance Design

To ensure high responsiveness, several performance-oriented design choices are included:

- **Serverless deployment** for automatic scaling
- **Caching layer with 1-hour TTL** for internship/job search results
- **React Server Components** for faster initial load
- **Reusable components** for rendering efficiency
- **Background processing with Inngest** for long-running AI tasks

These design decisions reduce latency, optimize cloud cost, and improve user experience.

4.9. Algorithmic Design

The system incorporates multiple internal algorithms:

a) Weighted Scoring Algorithm

Used to rank career recommendations based on:

- technical skills
- interests
- experience
- psychological attributes

b) Chain-of-Thought Prompting Logic

Used for roadmap and advanced content generation by guiding Gemini through stepwise reasoning.

c) Progression Difficulty Logic

Used in mock interviews to increase or decrease question difficulty depending on the user's performance.

4.10. File Structure Design

The internal codebase is organized as follows:

- **app/** – handles routes and page structure
- **actions/** – stores server-side logic such as assessment, roadmap, and internships
- **components/** – contains reusable UI parts
- **lib/** – utility files such as Prisma and Gemini integrations
- **prisma/** – contains schema and database design

4.11. UML Diagrams

4.11.1 Use Case Diagram Theory

The Unified Modeling Language (UML) serves as the indispensable architectural blueprint for the system's complex design. The Use Case Diagram for AI Based Career and Skill Recommendation System represents the absolute high-level functional scope of the entire ecosystem. The Primary Human User (Job Seeker) interacts simultaneously with several distinct operational sub-systems: the Assessment Sub-system responsible for diagnostic career pathing and capability evaluation; the Application Content Sub-system responsible for generating optimized resumes and tailored cover letters; the Interactive Practice Sub-system responsible for executing stress-tested mock interviews and rendering roadmaps; and the External API Sub-system which is the invisible layer where Clerk (Identity) and Gemini (Intelligence) operate to augment the platform's native capabilities.

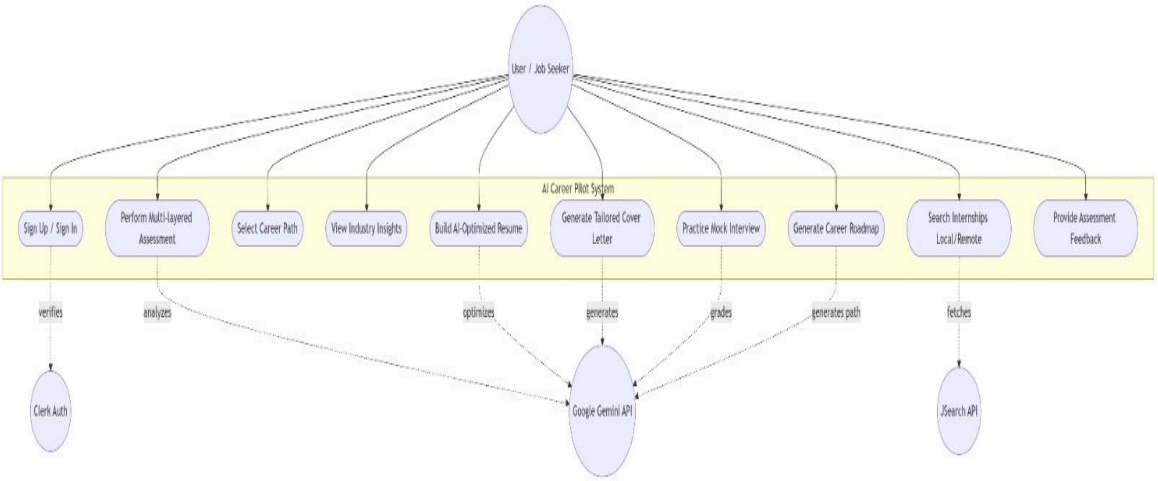

Fig. 4.1: Use Case UML Diagram

4.11.2 Activity Diagram

The highly detailed Activity Diagram traces the complete, uninterrupted lifecycle of a user session. It provides exceptional granularity in detailing the "Feature Selection Choice" stage, illustrating how a user can seamlessly diverge into any of the core generative modules from the central analytics dashboard. Furthermore, the diagram explicitly illustrates internal computational "Looping Constraints" utilized for iterative

mock interview questions, tracing how a user submits an answer, waits for AI evaluation, and then seamlessly proceeds to the next dynamically prompted question based on their performance vector. It also details the crucial "Conditional Logic Branches" that dictate whether the system queries the internal cache or reaches out to external APIs for internship data.

4.11.3 Sequence Diagram

The Sequence diagram visually demonstrates the comprehensive, end-to-end communication flow across the system's entire technology stack. It maps out the rigorous chronological interactions required to process a core user request—from the initial landing and Clerk Authentication JWT validation all the way through to the final AI generation phase. Most importantly, it clearly illustrates the secure intermediary role of the Next.js Server Actions, which receive unstructured client requests, securely query the Prisma/Neon DB for historical user context, and subsequently package that context into an engineered prompt for the Google Gemini API. Finally, the diagram traces how the AI's generated response is both permanently persisted back into the database and seamlessly rendered on the frontend Next.js interface for the user.

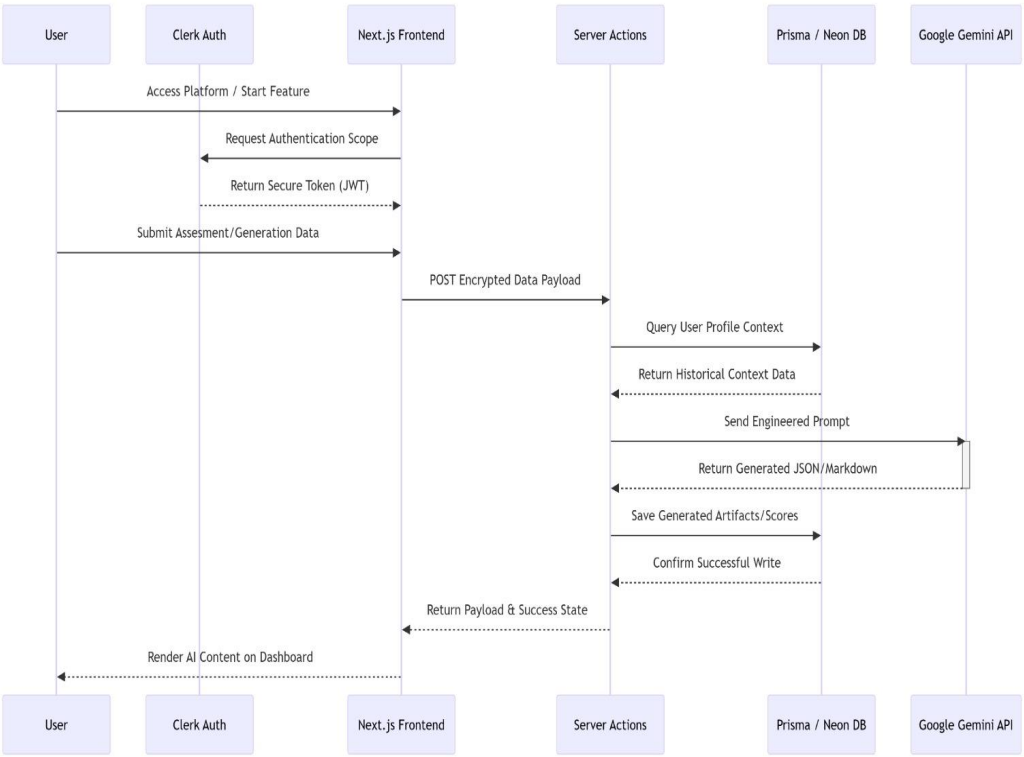

Fig. 4.2: Sequence UML Diagram

4.11.4 Class Diagram Theory

The Class Diagram is a foundational component of UML that strictly delineates the static application structure and the precise Object-Relational mapped relationships required for data integrity. In the AI Based Career and Skill Recommendation System architecture, the Class Diagram rigorously maps out the complex internal schema of the Neon PostgreSQL database. It illustrates pivotal cardinal relationships, such as how the centralized `User` class acts as the hub connecting in a one-to-many relationship to the `Resume` and `InterviewQuiz` classes. It meticulously ensures referential data integrity by demonstrating how foreign keys and primary keys restrict behavior, providing clarity on how the Prisma ORM correctly models complex, nested relational data without duplication or corruption.

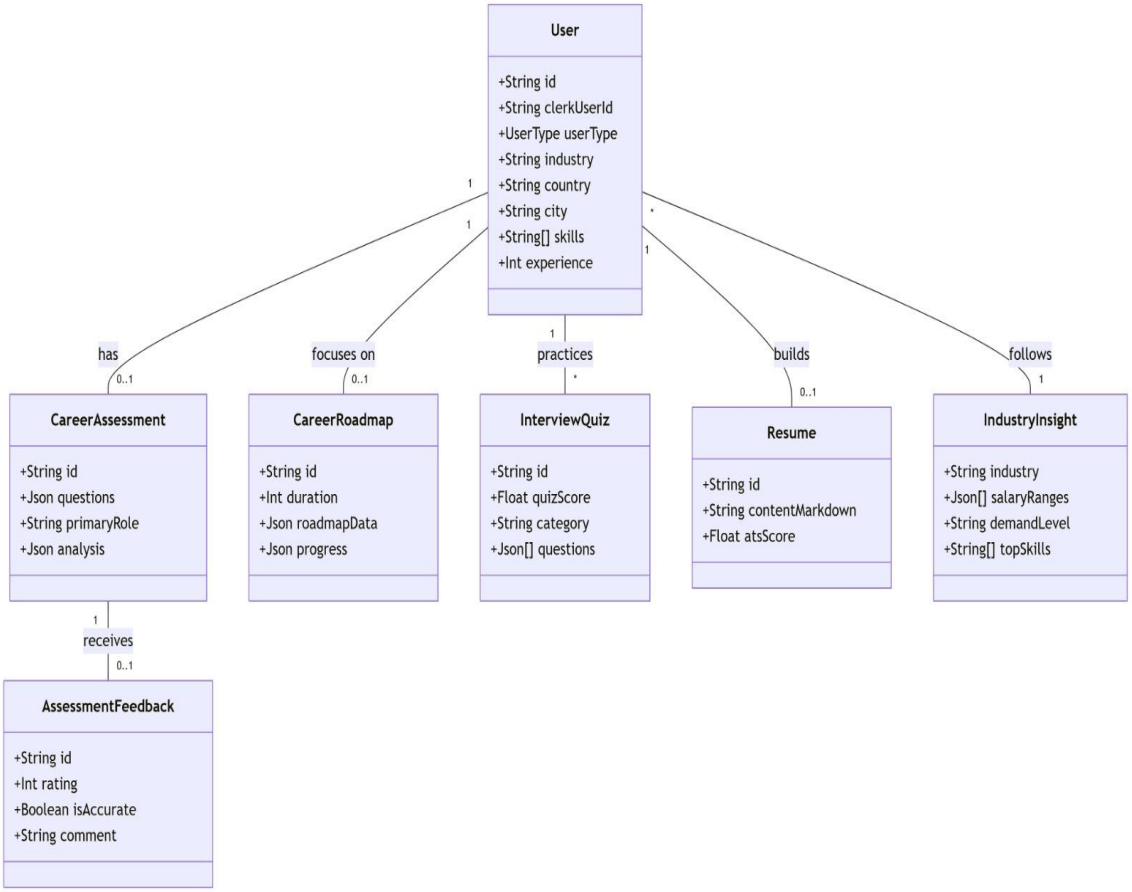

Fig. 4.3: Class UML Diagram

CHAPTER 5

IMPLEMENTATION

5.1 Methodologies

The development of the AI Based Career and Skill Recommendation System follows the **Agile Software Development Methodology**, which emphasizes iterative development, continuous improvement, and flexibility.

Agile Approach Used:

- The project is divided into **multiple development sprints**
- Each sprint focuses on building specific modules such as:
  - Career Assessment
  - Resume Builder
  - Mock Interview System
  - Job Search Module

Steps Followed:

1. **Requirement Analysis:** Understanding user needs such as career guidance, resume optimization, and interview preparation.
2. **Design Phase:** Creating system architecture, database schema, and UI layouts.
3. **Development Phase:** Implementing modules using modern frameworks and integrating AI APIs.
4. **Testing Phase:** Conducting unit testing, integration testing, and performance testing.
5. **Deployment Phase:** Deploying the application using serverless infrastructure.
6. **Maintenance:** Continuous updates, bug fixes, and feature enhancements.

Advantages of Agile in This Project:

- Quick adaptation to changes
- Faster feature delivery
- Continuous user feedback integration
- Improved system quality

5.2 Algorithms

The AI Based Career and Skill Recommendation System uses multiple intelligent algorithms to provide personalized outputs:

1. Generative Intelligence Algorithm (Chain-of-Thought)

- Breaks complex tasks into multiple steps
- Used for roadmap generation and resume creation
- Ensures accurate and structured outputs

2. Weighted Scoring Algorithm

- Assigns weights to user inputs such as:
  - Skills
  - Experience
  - Interests
- Calculates best career matches
- Outputs top recommended roles

3. Progression Difficulty Algorithm

- Used in mock interview module
- Adjusts question difficulty based on user performance

- Simulates real interview conditions

4. Temporal Caching (TTL) Algorithm

- Stores job search results for a fixed time (1 hour)
- Avoids repeated API calls
- Improves speed and reduces cost

Algorithm	Purpose
Chain-of-Thought	Structured AI output
Weighted Scoring	Career recommendation
Difficulty Algorithm	Interview level adjustment
TTL Caching	Improve performance

Table 5.1: Algorithms Used in the System

5.3 System Implementation Details

The implementation of AI Based Career and Skill Recommendation System is carried out using a **modern full-stack architecture**.

Frontend Implementation:

- Built using React-based framework
- Provides interactive UI components
- Handles user input and displays results

Backend Implementation:

- Server-side logic implemented using server actions
- Handles:
  - AI requests
  - Data processing

- API integrations

Database Implementation:

- Relational database used for storing:
  - User profiles
  - Resumes
  - Assessment data
  - Roadmaps
- Managed using ORM for efficient queries

AI Integration:

- AI model processes:
  - Career recommendations
  - Resume generation
  - Interview questions
- Uses structured prompts for accurate results

Key Features Implemented:

- Career assessment engine
- Resume and cover letter generator
- Mock interview system
- Job and internship search
- Dashboard for user interaction

5.4 Sample Code

Below is a sample implementation representing core logic used in the system:

1. **AI Career Discovery Engine Algorithm**

This module analyzes skills, interests, and personality traits to compute weighted career matches.

TypeScript

```
// lib/ai/careerDiscoveryEngine.ts

type Profile = {
  skills: string[];
  interests: string[];
  personality: string[];
};

type CareerScore = {
  career: string;
  score: number;
};

const CAREER_MATRIX: Record<string, string[]> = {
  "Software Engineer": ["coding", "problem solving", "algorithms"],
  "Data Scientist": ["statistics", "python", "machine learning"],
  "Product Manager": ["communication", "strategy", "leadership"],
  "UX Designer": ["creativity", "design", "psychology"],
  "Cybersecurity Analyst": ["networking", "security", "analysis"]
};

export function runCareerDiscovery(profile: Profile): CareerScore[] {
  const scores: CareerScore[] = [];

  Object.entries(CAREER_MATRIX).forEach(([career, traits]) => {
    let score = 0;
```

```
    profile.skills.forEach(skill => {
      if (traits.includes(skill.toLowerCase())) score += 3;
    });

    profile.interests.forEach(interest => {
      if (traits.includes(interest.toLowerCase())) score += 2;
    });

    profile.personality.forEach(trait => {
      if (traits.includes(trait.toLowerCase())) score += 1;
    });

    scores.push({ career, score });
  });

  return scores.sort((a, b) => b.score - a.score);
}
```

Explanation (for docs):

Uses weighted scoring

Compares user traits with career matrices

Returns ranked career paths

2 AI Generation of Career Paths

This generates structured career roadmaps dynamically.

TypeScript

```
// app/api/career-paths/route.ts

import { NextResponse } from "next/server";

export async function POST(req: Request) {
  const { career } = await req.json();
```

```
  const careerRoadmap = {
    "Software Engineer": [
      "Learn Programming Fundamentals",
      "Master Data Structures & Algorithms",
      "Build Full Stack Projects",
      "Contribute to Open Source",
      "Apply for Software Engineering Roles"
    ],
    "Data Scientist": [
      "Learn Python & Statistics",
      "Study Machine Learning",
      "Work on Data Projects",
      "Participate in Kaggle Competitions",
      "Apply for Data Science Internships"
    ]
  };

  const path = careerRoadmap[career] || [
    "Research the career",
    "Develop required skills",
    "Build portfolio projects",
    "Apply for internships"
  ];

  return NextResponse.json({
    career,
    roadmap: path
  });
}
```

Explanation:

API route generating structured career progression

Dynamic career roadmap generation

Easily extendable

3. Internship Fetching from External API

This service fetches internships from an external provider.

TypeScript

```
// lib/services/internshipFetcher.ts

const INTERNSHIP_API = "https://api.internships.example.com/search";

export async function fetchInternships(profile: {
  skills: string[];
  location: string;
}) {

  const query = new URLSearchParams({
    skills: profile.skills.join(","),
    location: profile.location
  });

  const res = await fetch(${INTERNSHIP_API}?${query}, {
    headers: {
      "Authorization": Bearer ${process.env.INTERNSHIP_API_KEY}
    },
    next: { revalidate: 3600 }
  });

  if (!res.ok) {
    throw new Error("Failed to fetch internships");
  }

  const data = await res.json();

  return data.results.map((internship: any) => ({
    title: internship.title,
    company: internship.company,
```

```
    location: internship.location,
    applyLink: internship.url
  }));
}
```

Explanation:

Calls external internship APIs

Uses caching (revalidate)

Filters results based on user profile

4 AI Career Recommendation using Gemini

Uses Gemini LLM to generate intelligent recommendations.

TypeScript

```
// app/api/ai-recommendation/route.ts

import { GoogleGenerativeAI } from "@google/generative-ai";
import { NextResponse } from "next/server";

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY!);

export async function POST(req: Request) {
  const { profile } = await req.json();

  const model = genAI.getGenerativeModel({
    model: "gemini-pro"
  });

  const prompt = `
  You are an AI career advisor.
  Analyze the student profile below and recommend 3 suitable careers.

  Skills: ${profile.skills.join(", ")}
  Interests: ${profile.interests.join(", ")}
  Personality: ${profile.personality.join(", ")}
```

CHAPTER 6

TESTING

6.1 Types of Testing

Testing is a critical phase in software development to ensure that the system is free from errors, performs efficiently, and meets user requirements. The AI Based Career and Skill Recommendation System undergoes multiple levels of testing to guarantee reliability and performance.

1. Unit Testing

Unit testing focuses on testing individual components or functions of the system.

- Testing individual modules such as:
  - Career assessment logic
  - Resume generation function
  - API request handlers
- Ensures each function works correctly in isolation

2. Integration Testing

Integration testing verifies the interaction between different modules.

- Testing communication between:
  - Frontend and backend
  - Backend and database
  - Backend and AI APIs
- Ensures smooth data flow across the system

3. System Testing

System testing evaluates the complete system as a whole.

- Tests full workflows such as:
  - User registration → assessment → recommendation
  - Resume creation → job application
- Ensures the system meets all functional requirements

4. End-to-End (E2E) Testing

E2E testing simulates real user scenarios.

- Verifies complete user journey:
  - Login → dashboard → feature usage → logout
- Ensures all modules work together seamlessly

5. Performance Testing

Performance testing checks system speed and responsiveness.

- Measures:
  - Page load time
  - API response time
- Ensures system works efficiently under load

6. Security Testing

Security testing ensures protection of user data.

- Validates:
  - Authentication system
  - Data encryption
  - Access control mechanisms

7. Usability Testing

Usability testing evaluates user experience.

- Checks:
  - Ease of navigation
  - Interface clarity
  - User satisfaction

6.2 Test Cases

Below are sample test cases used to validate the system functionality:

Test Case 1: User Registration

Test Case ID	**TC01**
Test Scenario	User Registration
Input	Valid username, email, password
Expected Output	User account created successfully
Result	Pass

Table. 6.1: Test Case 1-User Registration

Test Case 2: User Login

Test Case ID	**TC02**
Test Scenario	User Login
Input	Valid credentials
Expected Output	Login successful, redirect to dashboard
Result	Pass

Table. 6.2: Test Case 2-User Login

Test Case 3: Career Assessment

Test Case ID	**TC03**
Test Scenario	Career Recommendation
Input	User questionnaire responses
Expected Output	AI-generated career suggestions
Result	Pass

Table. 6.3: Test Case 3-Career Assessment

Test Case 4: Resume Generation

Test Case ID	**TC04**
Test Scenario	Resume Creation
Input	User details and experience
Expected Output	ATS-friendly resume generated
Result	Pass

Table. 6.4: Test Case 4-Resume Generation

Test Case 5: Mock Interview

Test Case ID	**TC06**
Test Scenario	Interview Simulation
Input	User answers
Expected Output	AI feedback and next question
Result	Pass

Table. 6.5: Test Case 5-Mock Interview

Test Case 6: Job Search

Test Case ID	**TC07**
Test Scenario	Job Listing
Input	Location and preferences
Expected Output	Relevant job/internship results
Result	Pass

Table. 6.6: Test Case 6-Job Research

Test Case 7: Performance Check

Test Case ID	**TC08**
Test Scenario	System Performance
Input	Multiple user requests
Expected Output	Fast response without lag
Result	Pass

Table. 6.7: Test Case 7-Performance Check

CHAPTER 7

RESULTS

7.1 Overview of Results

The AI Based Career and Skill Recommendation System was successfully developed and implemented as a fully functional, AI-driven career guidance platform. The system effectively integrates multiple modules such as career assessment, resume generation, mock interview simulation, and job search into a unified ecosystem.

The results demonstrate that the system is capable of delivering **accurate, personalized, and real-time career guidance**, significantly improving the efficiency and effectiveness of career planning for users.

7.2 Functional Results

The system was tested across all major modules, and the following outcomes were observed:

1. Career Assessment Module

- Successfully analyzes user inputs (skills, interests, experience)
- Generates top career recommendations with reasoning
- Provides personalized and relevant career paths

2. Resume Generation Module

- Produces **ATS-friendly, structured resumes**
- Highlights user achievements and skills effectively
- Reduces manual effort in resume creation

3. Cover Letter Generation

- Generates job-specific cover letters
- Aligns user profile with job requirements

- Improves application quality

4. Skill Gap Analysis and Roadmap

- Identifies missing skills accurately
- Generates **3-month, 6-month, and 12-month learning plans**
- Provides clear and actionable career development steps

5. Mock Interview System

- Simulates real interview environment
- Provides dynamic questions based on user performance
- Generates feedback and improvement suggestions

6. Job and Internship Search

- Fetches relevant job listings based on location
- Provides both local and remote opportunities
- Fast response due to caching mechanism

7.3 User Experience Results

The system provides a **smooth and user-friendly experience**:

- Easy navigation through dashboard
- Simple input forms for user data
- Clear and understandable outputs
- Responsive design for different devices

  Users can seamlessly move between modules without confusion, improving overall usability.

7.4 Accuracy and Efficiency

- AI-generated recommendations are **highly relevant and personalized**
- Resume and cover letter outputs align with modern hiring standards
- Interview feedback helps users improve performance
- Learning roadmaps are structured and goal-oriented

This demonstrates that the system effectively reduces the gap between user capabilities and industry requirements.

7.5 Comparative Results

Compared to traditional systems:

Feature	**Existing Systems**	**AI Based Career and Skill Recommendation System**
Career Guidance	Limited	AI-driven, personalized
Resume Creation	Template-based	ATS-optimized, dynamic
Interview Practice	Limited/paid	AI-based, adaptive
Integration	Fragmented	Unified system
Performance	Moderate	Fast and optimized

Table 7.1: Comparative Analysis of Existing System and Proposed System

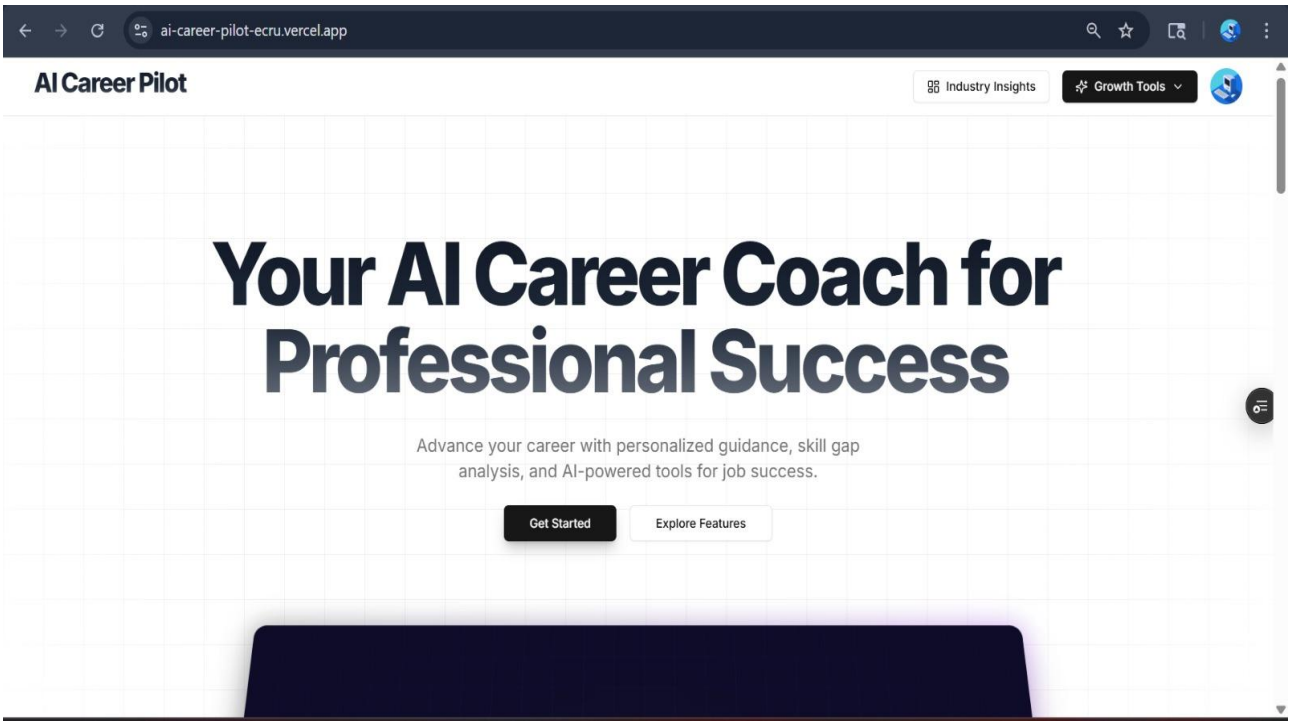

Fig. 7.1: Landing Page

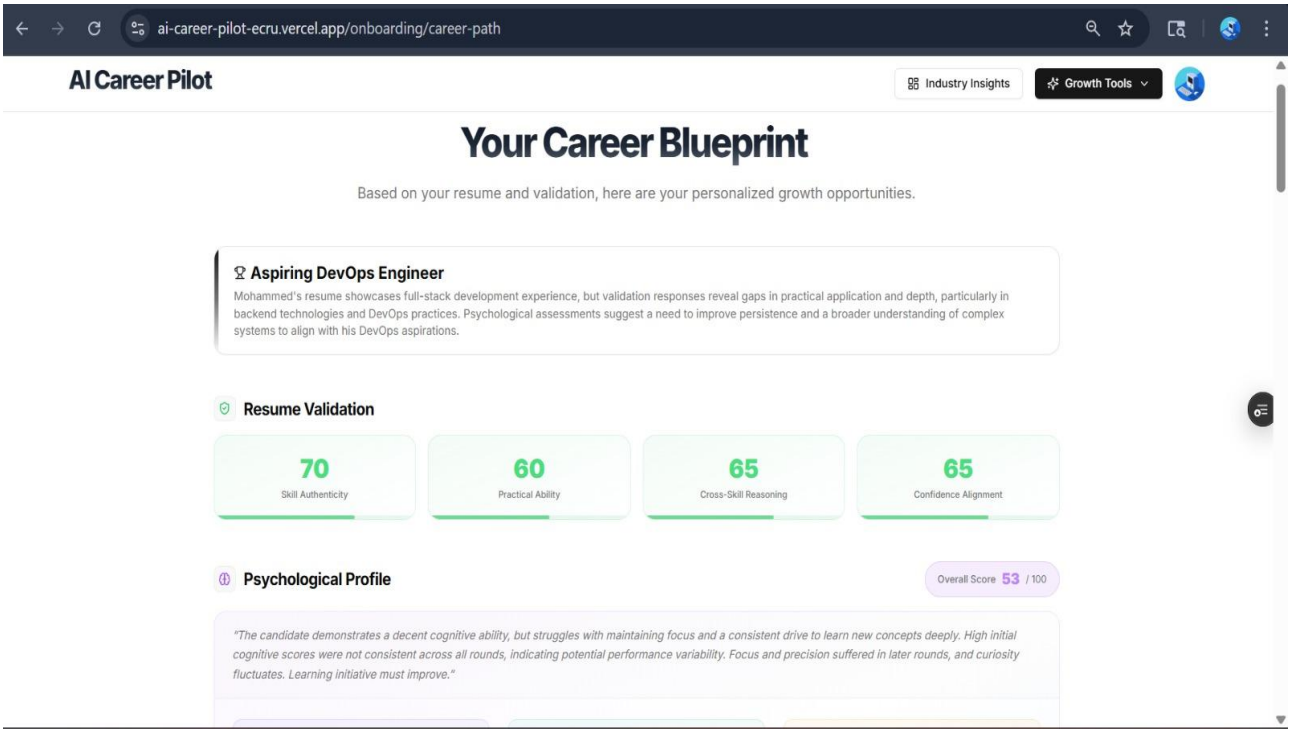

Fig. 7.2: Career Result Page 1

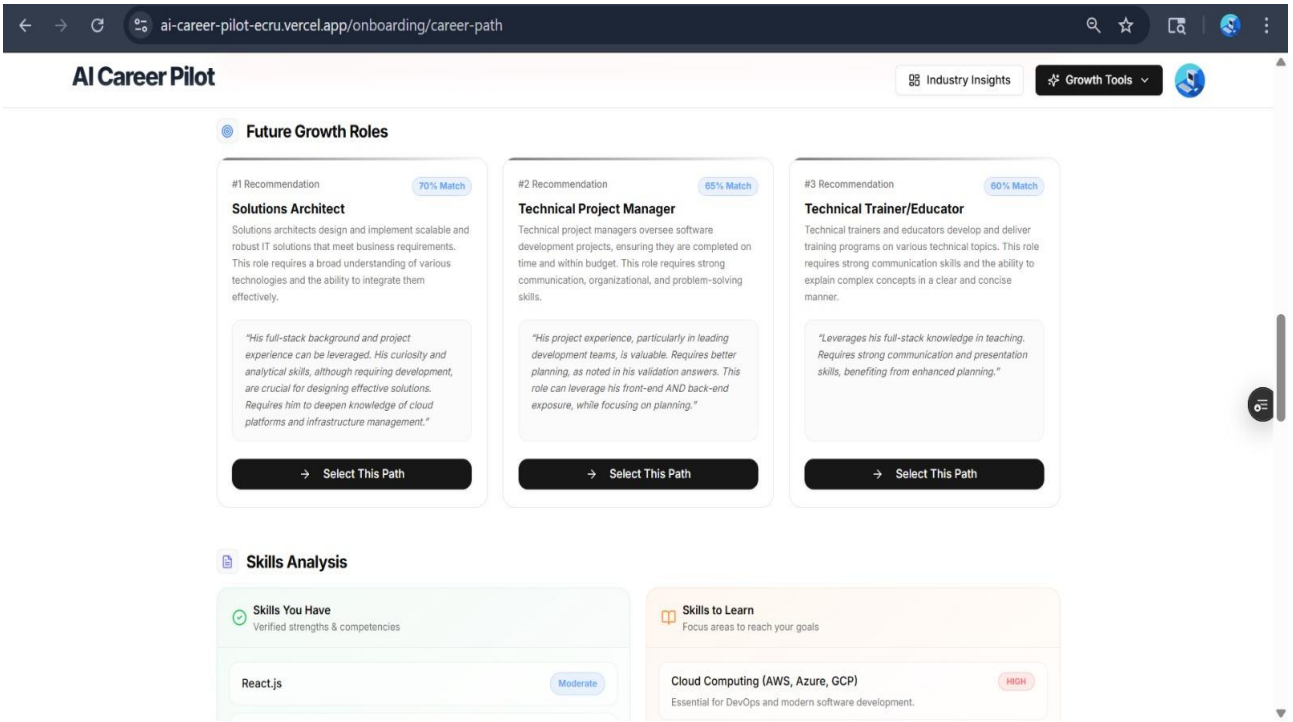

Fig. 7.3: Career Result Page 2

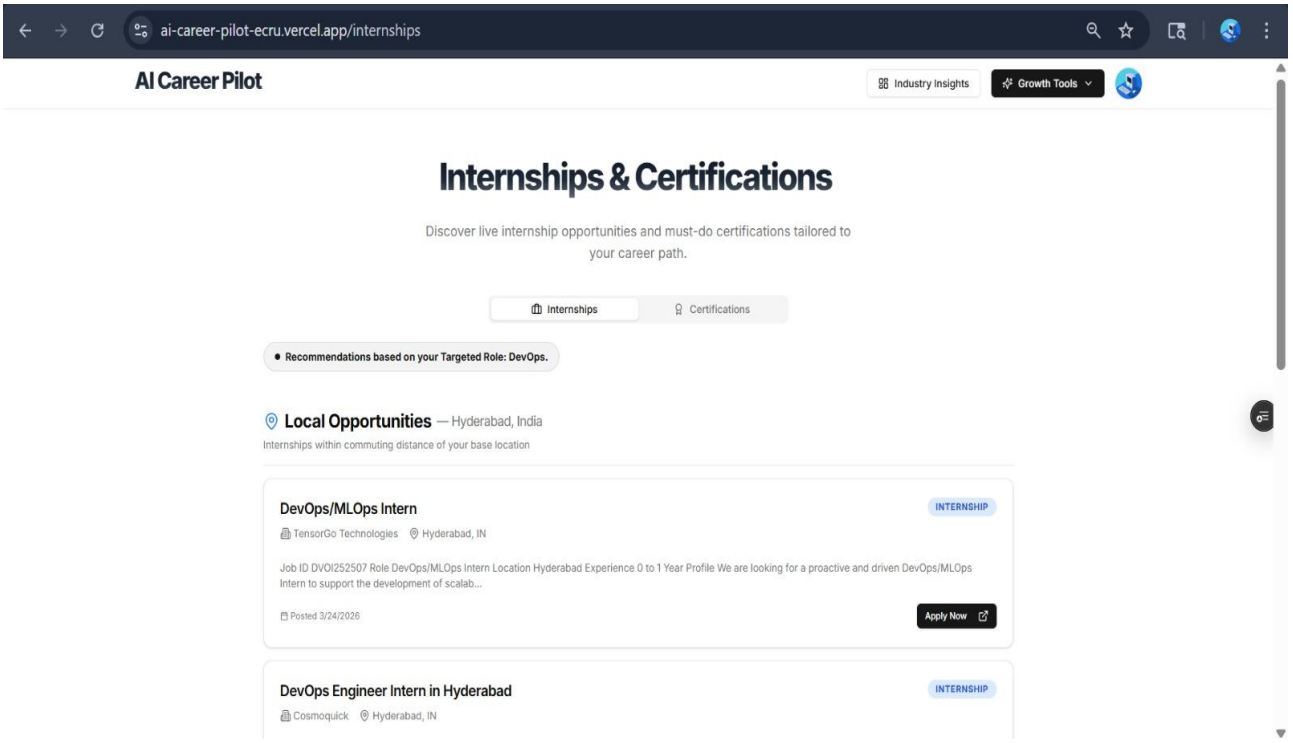

Fig. 7.4: Internship Recommendation

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The AI Based Career and Skill Recommendation System represents a significant advancement in the domain of career guidance and professional development. The project successfully addresses the major challenges faced by job seekers, including lack of proper career direction, ineffective resume presentation, and inadequate interview preparation.

By integrating advanced Artificial Intelligence techniques with modern web technologies, the system provides a **comprehensive, intelligent, and user-centric solution** for career planning. Unlike traditional systems that operate in isolation, AI Based Career and Skill Recommendation System unifies multiple functionalities such as career assessment, resume generation, mock interviews, and job search into a single platform.

The system demonstrates the effective use of AI in analyzing user profiles, identifying skill gaps, and generating personalized recommendations. It significantly reduces manual effort, improves decision-making, and enhances the chances of securing suitable job opportunities.

Furthermore, the implementation of a scalable and secure architecture ensures that the system performs efficiently under varying workloads. The use of caching mechanisms and serverless infrastructure enhances performance while reducing operational costs.

Overall, the project achieves its objectives by providing a reliable, efficient, and innovative solution that bridges the gap between user capabilities and industry requirements.

8.2 Future Scope

Although the AI Based Career and Skill Recommendation System is highly functional, there are several opportunities for future enhancements:

1. Voice-Based Interaction

- Integration of speech recognition and voice input
- Real-time conversational mock interviews

2. Video Interview Simulation

- AI-based facial expression and body language analysis
- Real-time feedback on communication skills

3. Integration with Professional Platforms

- Direct integration with job portals and professional networks
- One-click job application features

4. Advanced Analytics Dashboard

- Visual representation of user progress
- Performance tracking and improvement insights

5. Mobile Application Development

- Development of Android and iOS applications
- Improved accessibility for users

6. Real-Time Market Insights

- Integration of live industry trends and salary analysis
- Dynamic career recommendations based on market demand

7. Personalized AI Career Mentor

- Continuous guidance through chat-based AI assistant
- Long-term career tracking and suggestions

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